Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	02 March 2026
Team ID	699d5f7bfdf409fdbd1bebf5
Project Name	ResearchHub AI: Intelligent Research Paper Management and Analysis System using Agentic AI
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Registration	Registration through Email & Password form
		Registration through Google account
		Registration through Institutional login (University email)
FR-2	User Confirmation	Confirmation via Email verification link
		Confirmation via OTP verification
FR-3	User Login & Authentication	Login using registered email & password
		Secure login using OAuth (Google login)
		Password reset via email
FR-4	Workspace Management	Create multiple research workspaces
		Rename or delete workspace
		Assign research papers to workspace
FR-5	Research Paper Search	Search papers using keywords
		Retrieve results from ArXiv API
		Retrieve results from Semantic Scholar API
FR-6	Paper Import & Storage	Import research papers into workspace
		Upload PDF papers
		Extract text from uploaded PDFs
FR-7	AI Research Assistant	Ask questions about research papers
		Generate summaries of papers
		Compare methodologies across papers
FR-8	Research Knowledge Management	Save AI-generated insights
		Maintain conversation history
		Organize notes within workspace
FR-9	Paper Relationship Analysis	Identify connections between research papers
		Highlight key contributions and findings
FR-10	User Dashboard	View all workspaces and imported papers
		Manage research projects and conversations

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	The system must provide an intuitive and easy-to-navigate user interface so that researchers can easily search, organize, and analyze research papers.
NFR-2	Security	Secure authentication using JWT, encrypted password storage, HTTPS communication, and role-based access control to protect user data and

		research content.
NFR-3	Reliability	The system must consistently provide accurate responses from the AI model and ensure stable operation with minimal system failures.
NFR-4	Performance	Search results should appear within 2 seconds, and AI-generated insights should be returned within 4 seconds for typical queries.
NFR-5	Availability	The platform should maintain at least **99% uptime**, ensuring that researchers can access the system at any time.
NFR-6	Scalability	The architecture must support increasing numbers of users and research documents using scalable cloud infrastructure and vector databases.